KATTINTI SIVA KRISHNA
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Profile Synopsis

Strategic Engineering Leader with over **15 years** of experience in driving innovation across Software, Firmware, SoC systems, Hardware, Software, Embedded Platforms, IoT, Automation and Generative AI. I am **PMP-certified**, with **7+ years** of a strong background in Technical leadership, People management, Program execution, Team Management and leading cross-functional teams. I am currently pursuing my Executive MBA from **IIM, Kozhikode**. I successfully managing 5 agile teams and a **30+ member** organization (including 15 on-rolls), boosting productivity by **10%**, justify each resource utilization and reducing project delays by **30%**. As part of fostering a culture of innovation, I secured **9 patents** individually and mentored teams to deliver 13 more patents.

As a Technical Leader, I am responsible in designing scalable technical architectures and developing internal tools that improved engineering efficiency / Operation excellence by **5–10%**. I bring strong technical leadership with domain expertise in Automotive, HVAC, Security and Consumer Electronics. I demonstrated technical leadership in modern engineering excellence through automations, CI/CD adaption, implementing Generative AI solutions, and leveraging integrated tools like such as GitHub Copilot into engineering workflows. I come with expertise in aligning technology with business needs, converting business requirements into technical solution, driving platform-level innovation, and driving results in dynamic and evolving environments.

Employment Details

Since April 2022 with Stellantis as Engineering Manager
July 2018 to March 2022 Carrier Corporation as Senior Lead Engineer
May 2014 to July 2018 with Robert Bosch Engineering and Business Solutions Pvt. Ltd. as Sr. Embedded Developer
September 2012 to May 2014 with Lamtronics Ltd. as Embedded Software Developer
Jun 2011 to Aug 2012 with ACTS Techno Solutions Ltd. as Embedded Software Developer

Highlights:

- ✓ Proactively identified training gaps and ensured optimal resource alignment, placing the right talent in the right roles for maximum output.
- ✓ As Technical Manager, Created and deployed executive dashboards to track sprint burn-down, team velocity, and blocker resolution in agile, optimizing agile delivery cadence.
- ✓ Expertise in converting business problem into technical solution and present the pitch to executives, customers and get their early feedback.
- ✓ Expertise in people management, technical management, product management, vendor management along with team leadership
- ✓ Lead resource planning based on Product Increment features, Quarterly execution and identify cross functional dependencies, deliver quality solutions on time
- ✓ Establish team OKRs, cost optimizations and operational excellence to ensure reliable, performance and compliance
- ✓ Developed **IoT, AI/ML, Gen AI** projects proof of concepts, end-to-end products for cost saving and effective resource optimization.
- ✓ Led development and integration of crypto and security firmware protocols, including TLS / SSL and HSM based solutions w.r.t. OTA
- ✓ Led a team of engineers leveraging cryptographic algorithms like RSA, AES, SSH and industry standard APIs such as PKCS11 and open SSL for developing end-to-end IoT Products.
- ✓ Contributed in **18 end-to-end Products, 20 PoCs and 2 Products** using NPI process.
- ✓ Generated saving of **20%** energy expenses of system with help of sensors.
- ✓ Converted business problems / technical problems / product challenges into pragmatic solutions which includes converting to product features, technical designs and product strategy

- ✓ Collaborated cross-functionally with global stakeholders (**USA, EU, Germany, Singapore, China, and India**) to ensure seamless delivery and communication.
- ✓ Engaged in technical proposal development and cost estimations for **IoT-based** customer solutions, interfacing directly with customers, startups and Partner ecosystem with different models.

Core Skills

MANAGEMENT

People Management	**Technical Mentorship**	**Vendor Management**	**Team Leadership**	**Strategic Planning**
Process Management	**Change Management**	**Agile Methodologies**	**Culture Building**	**Conflict Resolution**
Resource Allocation	**Project Management**	**Decision Making**	**Communication**	**Cross Functional**
Result Orientation	**Team Building**	**Powerpoint/ EXCEL**	**MS Project**	

TECHNICAL

Software Architecture	**Technology Innovation**	**Emerging Technologies**	**Technology Research**
RTOS, Free RTOS, TI RTOS	**QNX, Linux/ Ubuntu**	**SoC, MCU, HAL**	**ARM Architecture**
Embedded Systems	**Andriod**	**Firmware Development**	**FOTA/SOTA/MOTA**
IoT, 2G & 4G, UWB	**Wi-Fi, BLE, LoraWAN**	**LTE, Sigfox**	**Sub-GHz**
Cloud (AWS, Azure, Google)	**AI & ML**	**Edge AI, AIoT, GEN AI**	**Tensorflow**
Snapdragon 6155, 7255,8255	**Docker**	**TCP/IP, UDP, LWM2M**	**Sensors, BSP**
Object Oriented Programming (OOP)		**Object-Oriented Design**	**Open-Source**

LANGUAGES

C | **Embedded C** | **C++(11,14 and above)** | **Python** | **Shellscript** | **Kotlin** | **Java**

TOOLS

JIRA	**Codebeamer** **RTC** **Azure Dev-Ops**	**Doors-DNG**	**RQM**
Github	**SVN** **VS Code Git-Co-Pilot**	**Makefile**	**GPT**
Eclipse	**Enterprise Architect (EA)**	**CI/CD (Jenkins, Teamcity)**	
Elastic	**Parasoft Dashboard** **Power BI**		

STANDARDS

ASPICE | **ISO26262** | **ISO22901** | **ISO14229-1**

SOFTWARE DEVELOPMENT METHODOLOGIES

Agile Development Methodologies | **Waterfall Development Methodologies**

CERTIFICATIONS

PMP®-PMI | **POPM-SaFe** | **Scrum Master-Scrum Alliances**

AI for Business Leaders Nano Degree | **Cybersecurity for Business Leaders Nano Degree**

X Culture Exposure and International Travel

- Managed vendor team of **Germany, Egypt** team members by providing SOW to deliver work package
- Managing global team with having reporting engineers in **USA, Europe** and **India**.
- As Innovation team member for the organization, worked collaboratively with **USA, China, Singapore** hubs and brought ~10 concepts out of **Product as a Service** model concept converted to Product.
- Visited **Germany** to collaborate on new wireless chip evaluation (**LoRa**), design, and development and testing.
- Visited **Germany** to collaborate with onsite partners and contributed to OTA system architecture design for IoT devices and successfully transitioning the work to India.

Innovation and Patents

- United States Patent (US20240362313A1) granted on **Dynamic switching of protocols for OTA updates**
- United States Patent (US20230141272A1) published on **Energy saving based on Dynamic Air Filtration**

- United States Patent (US20240362313A1) published on **Identifying vehicle security breach**
- United States Patent (US20240403019A1) published on **Secure Over the air updates btw multiple ECUs**
- United States Patent (US12327106B2) published on **Dynamic Switching of Protocols for OTA updates**
- France Patent (FR3157773A1) published on **Secure Communication between Vehicle & Infrastructure**
- Initiated and proved transitioned to teams **Requirement based Code development and Code coverage using LLM as part of** Team efficiency Improvement.
- Proven multiple LLM concepts to improve different system effective utilization
- Proposed and Converted Pay as a Service disruption model in HVAC domain

Foreign Languages

	Understanding	Speaking	Writing
English	**C1**	**C1**	**C1**
German	**A2**	**A2**	**A2**

Academia

2027 **MBA (Executive)** from **IIM, Kuzhikode (Pursuing)**
2011 B.Tech (ECE) from Sri Kottam Tulasi Reddy Memorial College of Engineering, AP, India with 68.5 %.

<u>*Refer to Annexure for Project Details</u>

ANNEXURE

Key Projects

Project: STLA Brain
Duration: Ongoing from April 2022
Description: **STLA Brain** is a cloud-integrated, service-oriented architecture that centralizes computing (HPC) and enables full over-the-air (OTA) access to Big Target ECUs, Small Target ECUs, Self-update, vehicle sensors and actuators by collecting data using inventory management service. By decoupling hardware and software cycles, it accelerates the delivery of new features without waiting for hardware updates. Developing software across different ECUs consist of **snapdragon 6155, 7255 and 8255** chipsets. This next-generation E/E architecture reduces system complexity, cutting the number of ECUs by half to approximately 60 and enables in-house development of new features in under six months, a quarter of the current traditional development time.
Team Size: 11
Position: Technical Engineering Manager
Environment: C++, Kotlin, Java, QNX, Linux, Qualcomm 8255, 7255, Parasoft, Cybersecurity, Android
Role:

- ✓ Responsible for developing and delivering quality software of Adaptive AUTOSAR architecture based OTA, & tested software components on SoC hardware of 6155, 7255 and 8255 with multiple environments including Linux, QNX and etc.,
- ✓ Proactively identify potential risks to the project deliverables and take appropriate measures to mitigate them to help minimizing impact on project timelines and budgets
- ✓ Identify training needs for the team members and develop training programs to ensure that they have the necessary skills and knowledge to perform their roles effectively
- ✓ Manage and lead a team of engineers to ensure the successful delivery of projects within the timeline and estimated budget.
- ✓ Collaborate with other cross functional software development teams, software architecture, system engineering, and quality assurance teams to ensure a seamless and integrated development process

- ✓ Ensure that OTA software on Adaptive Autosar architecture which meets relevant industry standards, regulations and guidelines
- ✓ Ensure optimal allocation of resources, including personnel, tools and equipment, to achieve project objectives.
- ✓ Continuously evaluate and improve the development process to ensure the highest levels of efficiency and quality.

Project: Efficient Co-Pilot Utilization
Duration: Ongoing from April 2025
Description: Utilizing Github Co-Pilot features as tool, this converts requirements into snippet C++ code, then from that code generating Static code violation free code, and provide html dashboard to provide how many requirements implemented in the code.
Team Size: 5
Position: Technical Engineering Manager
Environment: C++, QNX, Linux, Python, Github CoPilot, GPT, Tool
Role:

- ✓ Responsible for designing the concept and providing system architecture to deploy across cross functional teams
- ✓ Proactively identify potential risks to the project deliverables and take appropriate measures to mitigate them to help minimizing impact on project timelines and budgets
- ✓ Identify training needs for the team members and develop training programs to ensure that they have the necessary skills and knowledge to perform their roles effectively
- ✓ Collaborate with other cross functional software development teams, software architecture, system engineering, and quality assurance teams to understand their pain points, how our tool could provide solution to their pain-points
- ✓ Ensure optimal allocation of resources, including personnel, tools and equipment, to achieve project objectives.
- ✓ Continuously evaluate and improve the development process to ensure the highest levels of efficiency and quality.

Project: STLA Autodrive
Duration: Ongoing from April 2022
Description: **STLA Autodrive** integrates **STLA Brain** and **STLA Smart Cockpit** to deliver continuously evolving, intuitive, and reliable **ADAS (Advanced Driver Assistance System)** features that build driver confidence. **STLA AutoDrive**, developed in collaboration with BMW, aims to significantly extend hands-free driving time and distance. It supports **Level 2, 2+, and 3 autonomous driving**, with seamless OTA upgrades. The system offers both **eyes-off and fallback eyes-on driving modes** within a unified architecture. Eyes-off technology is expected to be integration-ready by year-end, with commercial deployment targeted for 2025.
Team Size: 8
Position: Technical Engineering Manager
Environment: C++, QNX, Linux, SOA, Qualcomm 8255, 7255, Parasoft, Cybersecurity
Role:

- ✓ Responsible for delivering quality software of Classic & Adaptive AUTOSAR of OTA & Diagnostics, tested & validated software components on multiple environments including Linux, QNX, QEMU and etc.,
- ✓ Proactively identify potential risks to the project deliverables and take appropriate measures to mitigate them to help minimizing impact on project timelines and budgets
- ✓ High Scalable platform which can be used across platforms
- ✓ Evaluate new technologies, tools and methodologies to determine their suitability for the project and make recommendations to senior management.
- ✓ Planning and allocating resources for current, maintenance, future projects, taking into account project timelines, budgets and resource availability
- ✓ Identify training needs for the team members and develop training programs to ensure that they have the necessary skills and knowledge to perform their roles effectively
- ✓ Manage and lead a team of engineers to ensure the successful delivery of projects within the timeline and estimated budget.

- ✓ Collaborate with other teams such as cross functional software development teams, software architecture, system engineering, and quality assurance to ensure a seamless and integrated development process
- ✓ Ensure that BSW OTA & Diagnostics software on both Classic & Adaptive Autosar which meets relevant industry standards, regulations and guidelines
- ✓ Ensure optimal allocation of resources, including personnel, tools and equipment, to achieve project objectives.
- ✓ Continuously evaluate and improve the BSW verification process to ensure the highest levels of efficiency and quality.
- ✓ Develop and Mentor team members to ensure that they have the necessary skills and knowledge to deliver high-quality work

Project: STLA Brain
Duration: April 2022 to August 2024
Description: **STLA Brain** is a cloud-integrated, service-oriented architecture that centralizes computing (HPC, 6155) and enables full over-the-air (OTA) access to Big Target ECUs, Small Target ECUs, Self-update, vehicle sensors and actuators, Cyber security, Diagnostics, Network Management . By decoupling hardware and software cycles, it accelerates the delivery of new features without waiting for hardware updates. This next-generation E/E architecture reduces system complexity, cutting the number of ECUs by half to approximately 60 and enables in-house development of new features in under six months, a quarter of the current traditional development time.
Team Size: 20+
Position: Engineering Manager
Environment: C++, QNX, Linux, SOA, Qualcomm 6255, Parasoft, Cybersecurity

Role:

- ✓ Responsible for delivering quality software of Classic & Adaptive AUTOSAR of OTA & Diagnostics, tested & validated software components on multiple environments including Linux, QNX, QEMU and etc.,
- ✓ Proactively identify potential risks to the project deliverables and take appropriate measures to mitigate them to help minimizing impact on project timelines and budgets
- ✓ Evaluate new technologies, tools and methodologies to determine their suitability for the project and make recommendations to senior management.
- ✓ Planning and allocating resources for current, maintenance, future projects, taking into account project timelines, budgets and resource availability
- ✓ Identify training needs for the team members and develop training programs to ensure that they have the necessary skills and knowledge to perform their roles effectively
- ✓ Manage and lead a team of engineers to ensure the successful delivery of projects within the timeline and estimated budget.
- ✓ Collaborate with other teams such as cross functional software development teams, software architecture, system engineering, and quality assurance to ensure a seamless and integrated development process
- ✓ Ensure that BSW OTA & Diagnostics software on both Classic & Adaptive Autosar which meets relevant industry standards, regulations and guidelines
- ✓ Ensure optimal allocation of resources, including personnel, tools and equipment, to achieve project objectives.
- ✓ Continuously evaluate and improve the BSW verification process to ensure the highest levels of efficiency and quality.
- ✓ Develop and Mentor team members to ensure that they have the necessary skills and knowledge to deliver high-quality work

Project: Low Cost IoT Gateway solution
Duration: Jul 2019 to March 2021
Description: Successfully converted business problem statement of high cost IoT gateway solution into low cost gateway solution. Challenge here is to understand high cost IoT solution limitations and advantages, then come-up with product fitment for this solution, technical deep dive with engineering teams, customer to understand which parameters important for customers to view,

which parameters required for OEM to provide predictive maintenance and other specific services to the customer. Derive technical solution which satisfy majority of the stakeholders including cost, mandatory parameters, new revenue generation services.

Team Size: 10
Environment: C, C++, QNX, Linux, Gateway
Position: Program Lead
Role:

- ✓ Requirements elicitation by interacting with the customers.
- ✓ Providing SRS (Software Requirements Specification), Architecture and other project and organization mandated documents for to startup and Engineering teams.
- ✓ Responsible for efforts estimation & budget estimations.
- ✓ Responsible for successful Product conversion along with Engineering Team.
- ✓ Collaborative working with other sub teams such as hardware, startup, engineering, cloud and mobile application teams to make sure that all the teams are in sync with each other for successful project delivery.
- ✓ Conducted reviews with stakeholders and made sure that the DoD (Definition of Done) is met for all the releases.

Project: Cooling as a Service disruption Model
Duration: Jul 2019 to March 2021
Description: Successfully converted business problem statement of service model for Air conditioners. I contributed to consider all the cooling parameters with IoT + AI solution which help to measure recurring model and provided Cooling as a Service model to customers under OPEX model.
Team Size: 15
Environment: C, C++, QNX, Linux, Gateway
Position: Program Lead
Role:

- ✓ Requirements elicitation by interacting with the customers.
- ✓ Providing SRS (Software Requirements Specification), Architecture and other project and organization mandated documents for to startup and Engineering teams.
- ✓ Responsible for efforts estimation & budget estimations.
- ✓ Responsible for successful Product conversion along with Engineering Team.
- ✓ Collaborative working with other sub teams such as hardware, startup, engineering, cloud and mobile application teams to make sure that all the teams are in sync with each other for successful project delivery.
- ✓ Conducted reviews with stakeholders and made sure that the DoD (Definition of Done) is met for all the releases.

Project: Disruptive Innovation solutions
Duration: Jul 2018 to March 2022
Description: Successfully demonstrated 10 technical PoCs which came as converting new technology evaluations / trends, new business opportunities, business problems, existing technical problems, and product challenges into pragmatic solutions, demonstrating to Directors, Executive board and Customers.
Team Size: 5
Environment: C, C++, QNX, Linux, Gateway, Alexa, Cybersecurity
Position: Program Lead
Role:

- ✓ Requirements elicitation by interacting with the customers.
- ✓ Providing SRS (Software Requirements Specification), Architecture and other project and organization mandated documents for Innovative solutions.
- ✓ Responsible for efforts estimation & budget estimations.
- ✓ Responsible for successful PoC demonstration.
- ✓ Collaborative working with other sub teams such as hardware, startup, engineering, cloud and mobile application teams to make sure that all the teams are in sync with each other for successful project delivery.

- ✓ Conducted reviews with stakeholders and made sure that the DoD (Definition of Done) is met for all the releases.

Project: Firmware development for BCDS IoT Edge devices (TDL, iRoll, Idefix, APLM, XDK, Backpack Demo)
Duration: 4.4 Years (May 2014 – June 2018)
Team Size: 10
Description: Responsible for understanding the requirements that comes from the Product Owner (PO) and providing the design, development and delivery of quality software. I worked as a Senior Developer for the following projects. Few of the IoT edge devices mentioned below are based on TI's CC2640 SoC which comes with an inbuilt BLE (Bluetooth Low Energy) interface, efm32 Giant Gecko controller along with Lora, cross development kit which can be used to develop IoT applications. Different sensors are interfaced to each of these devices based on the use case. All the following devices support FOTA (Firmware over the Air) update.

TDL (Transport Data Logger): TDL is an IoT edge device used for tracking the condition of a shipment during transit. It is mainly used for global logistics applications. It logs any shocks/tilts above configured threshold and a periodic environmental events such as temperature, humidity and pressure in an external memory. The device can be configured and logged events can be read using a mobile app through BLE interface.

iRoll: This device is used for predictive maintenance of roller mill. Once mounted in a roller, this device keeps monitoring the vibration and temperature of the roller and updates this data to a gateway through BLE interface. It can also measure the operation hours of a mill.

Idefix: This device is used for turning on/off, scheduling light functionality using mobile application. Smart phone over BLE is used to transmit command to IoT edge device, IoT edge device receives command from mobile, process it and then operates light according to received command.

APLM (Active Parking Lot Machine): This is IoT edge device is used to detect whether parking slot is available or not. It logs movement position / detection and periodic information such as radar, movement detection. The device can be configured and logged events can be read using either cloud / mobile application and data can be transferred through Lora interface from device to Gateway and from there to cloud.

XDK: This is IoT edge device / gateway used to develop rapid prototypes and demonstrate data in cloud. This device comes with temperature, light, humidity, accelerometer, gyrometer along with wireless capability of WiFi, BLE, Lora.

Backpack Demo: This device is used to detect someone trying to theft bag

Position: Senior Developer
Environment: C Programming on TI CC2640, efm32 Giant Gecko micro controllers using Eclipse IDE. Enterprise Architect tool used for design documentation and architecture diagrams, Cybersecurity
Role:

- ✓ Senior Developer for all the above projects
- ✓ Interacted with PO to understand the requirements.
- ✓ Worked along with Architect and provided software architecture for Fota, other packages using Enterprise Architect tool.
- ✓ Guided junior team members in completing the user stories.
- ✓ Reviewed the code and test cases written by the team members.
- ✓ Involved in the development of critical features.
- ✓ Made sure that the quality is maintained through continuous integration testing using Jenkins.
- ✓ Participated release reviews and made sure that the DoD (Definition of Done) is met for all the developed features.

Project: Firmware Development for Solar Inverter, Home automation
Duration: 1.8 Years (Sep 2012 – April 2014)
Description: Responsible for understanding requirement and designing according to requirement, developing firmware and testing. I worked as firmware developer for the following projects.

Solar Inverter: Understanding requirement and developing firmware which can help to get maximum of energy from solar panel. Motor with Light dependent sensor used to detect which direction Sun light is more accordingly solar panel will rotate, to get maximum intensity from sun. By using that solar panel generates energy and passes to battery to charge.

Home Automation: This is device used to automate appliances within premises. It logs events such as human movement detection, light intensity, temperature, Humidity, Access control within premises. According to Human motion, light intensity operates lighting within premises, human motion with temperature operates either Fan or Air conditioner within premises. Single responsible for developing complete development of this project.

Position: Embedded Developer
Team Size: 3
Environment: Languages: C
Role:

- ✓ Understanding requirement from customer
- ✓ Designing, developing and testing.
- ✓ Ownership of developing complete product development.
- ✓ Testing with all corner cases and made sure delivering bug free product.

Project: Firmware Development for Industrial Timer
Duration: 1.2 Years (June 2011 to Aug 2012)
Team Size: 3
Description: Responsible for understanding existing code, providing bug fixes and adding new features for the Industrial Timer.

Industrial Timer: This is device majorly used to save electricity of commercial office. Based on human detection, operating mains power supply, lights, fans, etc. Timer used to preset Main source power supply. Based on timer, mains power sources will work. During main power source usage, within premises validates motion detection and based on movement operates premises appliances.

Position: Embedded Developer
Environment: 8bit microcontroller, 16bit microcontroller
Role:

- ✓ Understanding the existing code base, fixing the issues if any and adding new features for the current processor.
- ✓ Understanding requirement from customer
- ✓ Designing, developing and testing.
- ✓ Ownership of developing complete product development.
- ✓ Testing with all corner cases and made sure delivering bug free product.